

When the best score is not the best complement

A / ORIGINAL AIRA-EVO

Absolute score controls the gate

Score-softmax selection · independent repair branches

RANKED BY VALIDATION SCORE

01 REPAIR BRANCH A AUC 0.94656
Draft error → Debug error → Debug

02 REPAIR BRANCH B AUC 0.85546
Draft timeout → Debug

SCORE-ONLY FINAL SELECTION

Branch A survives

Branch B contributes nothing to the final program.

FINAL DEBUG

VALIDATION AUC

0.94656

HELD-OUT AUC

0.94852

B / OPENMLE-EVO

Complementarity changes who gets selected

Score identifies strength; gain and novelty preserve useful alternatives

● PARENT A / SCORE LEADER

Deep ResNet-SE

64-Mel · AMP · test-time augmentation

AUC 0.99187

SCORE RANK #1

● PARENT B / GAIN LEADER

Log-Mel + temporal deltas

Delta and Delta-Delta channels

AUC 0.98773

SCORE #6 / GAIN #1

THREE-FACTOR PARENT SELECTION / SAME 10-PARENT POOL

SCORE / ABSOLUTE QUALITY

1.0

GAIN / PARENT IMPROVEMENT

0.6

NOVELTY / BRANCH DIVERSITY

0.3

P(SELECT PARENT B)

10.47%

17.09%

+63.2% RELATIVE

SELECTED IMPROVE CHILD

B's temporal channels

EMA · 5-way TTA · label smoothing

VALIDATION AUC

0.99203

HELD-OUT AUC

0.99386